

Numerical Methods in Chemical Engineering and Problem Solving

ChE 348 Fall 2023

Course Description: The goal of ChE348 is to teach students how to apply computational methodologies to solve engineering problems when no closed-form, analytical solution exists. Achievement of this goal requires learning the basics of structured programming as well as learning how to combine engineering knowledge, judgment, and intuition to develop reasonable approximations through the engineering modeling process. Because mathematical judgment and approximations are involved, the material in this course will be somewhat more open-ended than the material covered in other courses. Emphasis will be placed on understanding the basic concepts behind the various numerical methods studied, implementing basic numerical methods using the MATLAB structured programming environment, and utilizing more sophisticated numerical methods provided as built-in MATLAB functions. This approach is taken since understanding how numerical methods work is essential for choosing the correct method and understanding its limitations. At the same time, the existence of commercial numerical libraries makes it inefficient and unnecessary for students to re-develop complex existing numerical routines.

By the end of this course, you should be able to do the following:

Structured programming. Understand basic structured programming concepts involving decision making, loops, functions, and parameter passing implemented within the MATLAB programming environment.

Numerical methods. Understand the most common numerical methods used in engineering analysis, when to use each method, and how to implement basic methods in a structured manner using MATLAB's programming language.

Numerical accuracy. Estimate the amount of error inherent in different numerical methods.

Numerical efficiency. Assess the efficiency of a selected numerical method when more than one option is available to solve a certain class of problem.

Numerical stability. Understand the convergence properties and limitations of different numerical methods.

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Office phone: 471-4438

Office hours:

TA:

Lectures: TTh 9:30 – 11:00 AM

Please be punctual. Attendance is not compulsory, but is strongly recommended. Bring your laptop to all the lectures.

Recitation: M 1:00PM - 3:00PM

We will use MATLAB to solve problems. There will be a short quiz.

Optional Books:

[Numerical Methods with Chemical Engineering Applications](#) by Kevin

D.Dorfman and Prodromos Daoutidis, Cambridge University Press, 2017.

Introduction to Chemical Engineering Computing, by Bruce A. Finlayson, Wiley, 2012.

A Friendly Introduction to Numerical Analysis by Brian Bradie, Prentice Hall, 2006.

Problem Solving in Chemical and Biochemical Engineering with Polymath, Excel and Matlab, by Cutlip M. B., and Shacham M., 2nd Edition, Prentice Hall, 2008.

Final Grade

Homework (MATLAB problem solving)	20 %
Matlab problem solving Quiz (During recitation)	10%
Three Exams + Final Exam (in-person + Takehome)	70 %

Final letter grade will be given based on Class average. The class average for this course is B-. In this course +/- grading will be used.

Exam Dates: Exam 1: September 18th , 1 to 2:50 PM (During recitation)

Exam 2: October 16th , 1 to 2:50 PM (During recitation)

Exam 3: November 27th , 1 to 2:50 PM (During recitation)

Final Exam:

Absence: No excuse for missed exams. If you miss an exam due to health or family reason, you have to take a makeup exam that will be given on Dec 11th .

Homework: Homework will be assigned weekly and is due the following week. The MATLAB problems (code and output) should be copied into a word file and uploaded to CANVAS.

Duplicate assignment solutions are unacceptable even if you work together as a group. To be successful in exams and quizzes, try to work on the homework problems alone. Don't share or copy MATLAB code. If two identical MATLAB codes are found in the exam, quiz or homework, both students will be given zero and necessary action will be taken.

Course prerequisites: CHE 210, CHE 317 and M 427J or M 427K with a grade of C at least in each of them. Transport phenomena (shell balance models) knowledge is very useful in understanding the physics of the problems we will be solving in this class.

In the exams and homework, you will be given a detailed transport model (along with the necessary boundary and initial conditions). You should be able to solve these problems using MATLAB functions.

Instructions to submit the MATLAB homework, exam and quiz files for ChE 348 on Canvas

Copy all the matlab code and outputs in a word file.

Upload the word file

Name the word file in the following way.

File and Name Format

<EID of student>_<Assignment name>

Example: If your EID is abc123 and you want to submit the script files for assignment HW2 the name of the zipped folder should be

Homework **abc123_HW2**

The same is true for a quiz or an exam

Quiz **abc123_Quiz1**

Exam **abc123_exam1**

Sharing of Course Materials is Prohibited

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.